

USA Regional Sales Analysis

Technical Portfolio Report | Acme Co. | 2014–2018

\$1B+	4 Years	37.9%	\$150M	0.91
Total Sales	Data Span	Export Margin	Regional Gap	Price-Revenue Correlation

SECTION 01

Project Overview & Problem Statement

Acme Co. required actionable insights from its 2014–2018 USA sales data to identify key revenue drivers across products, channels, and regions. The core problem was a lack of consolidated visibility into seasonal trends, pricing anomalies, and margin profitability — which impeded the executive team's ability to optimize pricing and formulate effective market-expansion strategies.

"The USA Regional Sales Analysis project is an end-to-end business intelligence solution that transformed four years of raw, disconnected Excel sales data into an interactive Power BI dashboard. By leveraging Python for rigorous data wrangling and EDA, the project uncovered a highly profitable but underutilized export channel and quantified a \$150M regional performance gap, directly empowering stakeholders to optimize promotional strategies and drive data-backed growth."

SECTION 02

Tech Stack

Layer	Technology	Why Chosen
Data Source	Excel (.xlsx)	Ubiquitous format representing the client's initial raw business data storage.
Data Wrangling	Python (Pandas, NumPy)	Extremely efficient for cleaning, joining relational tables, and feature engineering.
Exploratory Analysis	Python (Matplotlib, Seaborn, Plotly)	Enables deep-dive statistical visualizations, correlation mapping, and temporal trend analysis.
Business Intelligence	Microsoft Power BI	Industry standard for interactive, dynamic, executive-facing visual dashboards.

SECTION 03

Project Architecture & File Structure

Regional Sales Dataset.xlsx	Raw data input — Orders, Customers, Regions sheets
EDA_Regional_Sales_Analysis.ipynb	Python data pipeline and exploratory analysis
Sales_data (EDA Exported).csv	Cleaned, feature-engineered output dataset
Background/	UI/UX elements for the BI dashboard
SALES REPORT.pbix	Interactive Power BI dashboard
PPT — Regional Sales Analysis.pptx	Stakeholder presentation deck

End-to-End Data Flow

[Raw Excel Data]

- Python / Jupyter (Wrangling & Feature Engineering)
- Denormalized CSV Export
- Power BI (Modeling & DAX Measures)
- Interactive Executive Dashboard

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Storage Design

The project utilises a static, flat-file **ELT (Extract, Load, Transform)** approach. Raw data resides in a multi-sheet Excel file representing isolated domains (Sales, Customers, Regions). This is extracted, joined into a unified denormalised schema using Pandas, and exported to a flat CSV file, which serves as the optimised data source for the BI visualisation layer.

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Core Logic / Data Processing Details

- **Data Cleaning:** Null values and irregular data types were rectified. Column headers were systematically lowercased for standardised syntax access.
- **Targeted Masking:** Budgets were explicitly masked (pd.NA) for non-2017 data points upon discovering that the company only maintained accurate budget tracking for that specific year — preventing skewed historical averages.
- **Feature Engineering:** Engineered critical business metrics including *total_cost* (Quantity × Unit Cost), *profit* (Revenue – Total Cost), and *profit_margin_pct* to enable analysis independent of volume.

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Key Technical Deep-Dive: Exploratory Data Analysis (EDA)

The Python EDA phase was critical for uncovering insights obscured by raw numbers. Extensive univariate, bivariate, and temporal analyses were conducted across the full dataset.

- **Outlier Detection:** Boxplots isolated extreme structural anomalies — high-end bulk orders and low-end promotional giveaways (\$0–\$100 SKUs).

- **Correlation Mapping:** Heatmaps revealed a near-perfect positive correlation (0.91) between Unit Price and Revenue, while order Quantity showed zero correlation — proving revenue growth is driven by item value rather than volume.

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Automation / Deployment

The data pipeline was executed sequentially within Jupyter. The resulting Power BI dashboard (*SALES REPORT.pbix*) is designed for desktop interaction or immediate deployment to **Power BI Service** for secure, organisation-wide, role-based sharing.

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Dashboard / UI Features

- **Tailored UI:** Utilised custom high-fidelity background templates for a professional executive look.
- **Key Metrics:** Total Revenue, Order Volume, and Profit Margins displayed as headline cards.
- **Regional Dominance:** Visual tracking of market share — West coast driving ~35% of total sales.
- **Channel Profitability:** Export channel leading at 37.9% margin despite representing only 15% of order volume.

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Setup & Running Instructions

```
# 1. Clone the repository
git clone [repository_link]
cd Sales-Analysis

# 2. Install Python dependencies
pip install pandas numpy matplotlib seaborn plotly jupyter

# 3. Execute the Data Pipeline
jupyter notebook EDA_Regional_Sales_Analysis.ipynb

# → Output: Sales_data(EDA Exported).csv will be generated

# 4. View Dashboard

# Open SALES REPORT.pbix in Microsoft Power BI Desktop

# Refresh data connections if prompted
```

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Challenges & Solutions

Challenge	Solution
Disjointed Data Entities	Utilised Pandas merge() functions to join distinct Orders, Customers, and Regional datasets into a single unified analytical dataframe.

Inconsistent Budget Tracking	Identified that budgets were only tracked for 2017. Used df.loc masking to blank non-2017 fields, preventing skewed historical baselines.
Skewed Margin Calculations	Used programmatic thresholding and boxplots to isolate massive bulk orders and cheap test SKUs — logically separating them to accurately calculate standard profitability.

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Limitations

- Data source is static — insights are limited strictly to the 2014–2018 timeframe.
- Flat-file architecture lacks a live database connection — manual pipeline reruns required for any new data drops.

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Future Improvements

- **Database Migration:** Move raw Excel storage into a robust relational database (e.g., PostgreSQL) for scalability.
- **Pipeline Automation:** Implement Apache Airflow to automate daily ETL ingestion scripts without manual intervention.
- **Predictive Analytics:** Integrate time-series forecasting (Prophet or ARIMA) to predict future sales cycles based on historical seasonality.

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Resume Bullet Points

- Spearheaded an end-to-end sales analytics ETL pipeline using Python (Pandas, NumPy) to process and join a 4-year regional dataset, engineering critical profit and cost features.
- Executed comprehensive Exploratory Data Analysis (EDA) across multi-dimensional tiers, uncovering a highly profitable export channel (37.9% margin) and a \$150M regional performance gap.
- Designed an interactive executive dashboard in Power BI visualising \$1B+ in total sales, highlighting West region market dominance driving 35% of total revenue.
- Formulated actionable, data-driven business recommendations by isolating a critical 2017 revenue anomaly (drop to \$21.2M) and defining optimal pricing strategies.

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Key Interview Q&A

Q: Why Python over Power Query for cleaning?

A: Pandas offers significantly more programmatic flexibility for complex cleaning, missing-value handling, and deep statistical EDA (correlation heatmaps, boxplots) — before data ever hits the visualisation layer.

Q: How did you handle the messy budget data?

A: During EDA I discovered budgets were only consistently recorded for 2017. I used Pandas to logically mask non-2017 values as pd.NA to ensure historical averages were not skewed by zeros.

Q: What was the most surprising insight?

A: Despite Wholesale dominating 54% of pure volume, the Export channel (only 15% volume) maintained the highest overall profit margins at ~37.9% — making it the primary recommendation for targeted expansion.

Q: How did you calculate profit margins?

A: I engineered a profit feature (Revenue – Total Cost) and calculated margin percentage. Scatter plots revealed dense, consistent margin bands (18%–60%) regardless of unit price — proving the baseline pricing multiplier was highly standardised across the catalogue.

Q: How would you scale this for real-time data?

A: Migrate flat-file storage to PostgreSQL, build an automated Python ETL script orchestrated via Airflow to ingest daily transactions, and connect Power BI directly via DirectQuery.

Q: Explain your data flow architecture.

A: Flat-file ETL pipeline: Raw multi-sheet Excel was ingested into Jupyter, transformed and joined with Pandas, exported as a denormalised CSV, and that clean CSV feeds the Power BI semantic model.

Q: What challenges did outliers present?

A: Extremely low-priced promotional items and massive high-value bulk orders severely skewed average unit prices and margin generalisations. Boxplots identified distinct thresholds for logical grouping, preventing dilution of core metrics.